

ER. PRASHANT PHUYAL

Associate AI Engineer | Data Science, ML, AI Enthusiast | Learner

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PROFESSIONAL SUMMARY

AI Engineer specializing in LLM applications, Retrieval-Augmented Generation (RAG), AI chatbots, and production-ready Generative AI systems. Experienced in building document-grounded AI assistants and AI features for healthcare, education, travel, recruitment, and home-services platforms. Skilled in end-to-end RAG pipelines including document ingestion, chunking, embeddings, vector search with Qdrant and FAISS, prompt engineering, and LLM integration using Gemini and OpenAI. Strong experience building scalable FastAPI services with PostgreSQL, MongoDB, Redis, and asynchronous Python. Promoted from AI/ML Intern to Associate AI Engineer at Dome Infosys.

TECHNICAL SKILLS

Generative AI & LLM: Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Chatbots, Voice AI Agents, LLM Integration, Intent Classification, Conversation Memory, Structured JSON Output, LangChain, LangGraph, Google Gemini API, OpenAI API, HuggingFace

Vector Search & Retrieval: Qdrant, FAISS, Embeddings, Semantic Search, Document Ingestion Pipelines (PDF, DOCX, TXT, Markdown, Excel), Text Chunking, Top-K Retrieval, sentence-transformers

Backend & APIs: FastAPI, Python, Pydantic, Asynchronous Python, REST API Design, Uvicorn, SQLAlchemy, Celery, JWT Authentication, Rate Limiting, GitHub Actions, Pytest, Swagger / OpenAPI, Git

Machine Learning & Computer Vision: TensorFlow, Keras, PyTorch, Scikit-learn, Faster R-CNN, YOLO, Object Detection, Vision LLMs, OpenCV, OCR (EasyOCR), Model Training & Evaluation

Databases & Data Analysis: PostgreSQL, MongoDB, Redis, SQL, Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA)

PROFESSIONAL EXPERIENCE

Dome Infosys | Kathmandu, Nepal

Associate AI Engineer (promoted from AI/ML Intern) | More than 1 Year

- Design, build, and deploy production RAG chatbots and Generative AI features for enterprise clients across healthcare, education, travel, recruitment, retail, and financial services.
- Delivered document-grounded AI assistants for Vatsalya IVF (Ziva), Babyloan School, YatriFly, Nepwood, Makalu, SewaFund, Rojina Beauty Salon, and Pokhara Motors.
- Own end-to-end RAG pipelines: document ingestion, chunking, embedding generation, Qdrant and FAISS indexing, top-K semantic retrieval, and grounded LLM response generation with domain-specific prompts.
- Build asynchronous FastAPI backends with conversation persistence in PostgreSQL and MongoDB, Redis caching and rate limiting, and automated deployment using GitHub Actions.

PROJECTS AT DOME INFOSYS

Ziva — IVF AI Assistant (Vatsalya IVF)

FastAPI, Gemini, OpenAI, Qdrant, LiveKit, PostgreSQL, SQLAlchemy, Alembic, Redis, GitHub Actions

- Built a domain-restricted IVF and fertility AI assistant supporting text and real-time voice interactions, using RAG and clinic-approved knowledge to provide grounded responses.
- Implemented the RAG pipeline — intent classification, Qdrant vector search, and IVF-constrained prompt generation — keeping the assistant accurate and in-scope on a sensitive medical topic.
- Developed the real-time voice agent with LiveKit, plus conversation persistence in PostgreSQL and Redis caching and rate limiting.

Jobscater — Generative AI Features

Python, FastAPI, LLM APIs, Prompt Engineering, Structured JSON Output

- Built the platform's Generative AI feature set across employer job posting, job-seeker guidance, and the AI CV Maker, delivered as backend APIs for the live web application.
- Developed LLM generation for job skills, keywords, roles & responsibilities, and job specifications, letting employers produce a complete structured job post from minimal input.
- Implemented CV Maker APIs for summary generation, skill suggestion, ATS scoring with actionable feedback, and interview-preparation questions generated from candidate CV data.

Banau.ai — AI Microservice

FastAPI, Gemini, OpenAI, LangChain, Qdrant, PostgreSQL, Redis, Celery

- Developed all AI features of a production AI microservice exposing nine AI domains — including chatbot, voice, image diagnosis, pricing, provider matching, and fraud detection — consumed over HTTP by separate CRM and Booking services.
- Built a bilingual Nepali/English/Hindi assistant with intent detection, service-category routing, multi-turn memory, and RAG answers retrieved from Qdrant.
- Implemented a multi-provider LLM setup with Gemini as primary and OpenAI as automatic fallback, and developed confidence-tiered image damage diagnosis using vision LLMs.

OTHER PROJECTS

Traffic Sign Detection System for Nepal — Trained an end-to-end Faster R-CNN model recognizing 9 classes of Nepalese traffic signs, covering data collection, annotation, augmentation, training, and evaluation. [Python, TensorFlow, Keras, OpenCV, FastAPI]

Nepali OCR Engine — Built a Devanagari and English OCR pipeline extracting structured text from scanned documents, using custom OpenCV preprocessing to improve recognition accuracy. [Python, EasyOCR, OpenCV]

KnowledgePilot — RAG & Agent API — Personal project building a document-grounded question-answering API with citations and a Streamlit chat UI. [FastAPI, LangChain, Qdrant, OpenAI, Streamlit]

Student Stress Factor Analysis — Performed exploratory data analysis and trained classification models to identify the strongest predictors of student stress. [Python, Pandas, Scikit-learn, Matplotlib]

EDUCATION

Cosmos College of Management & Technology | Kathmandu, Nepal

Bachelor in Computer Engineering | 2021 – 2025 | CGPA: 3.6 / 4.0

CERTIFICATIONS & LANGUAGES

Certification: Data Science with Python

Languages: Nepali (Native), English (Professional), Hindi (Conversational)